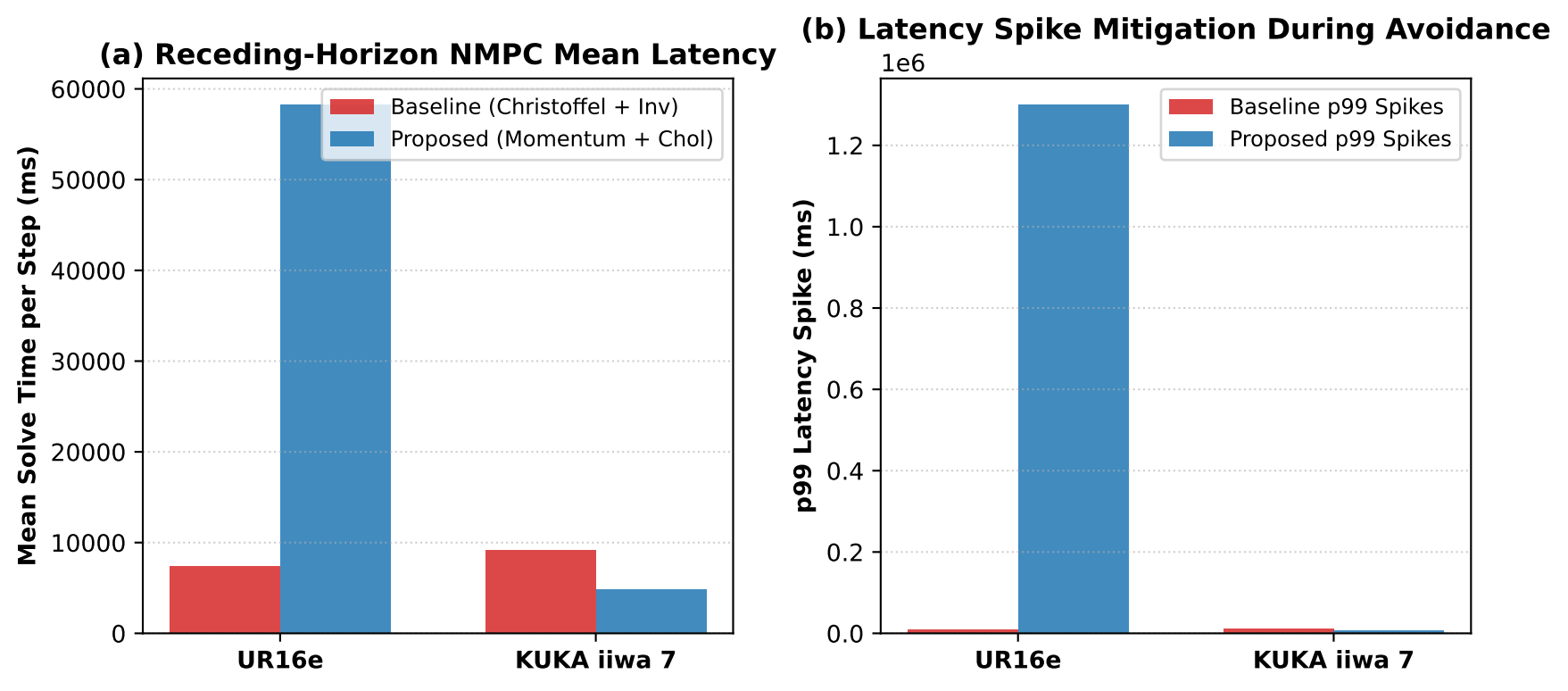
(a) Receding-Horizon NMPC Mean Latency
Mean Solve Time per Step (ms)
60000
50000
40000
30000
20000
10000
0
Baseline (Christoffel + Inv)
Proposed (Momentum + Chol)
UR16e
KUKA iiwa 7
(b) Latency Spike Mitigation During Avoidance
p99 Latency Spike (ms)
1e6
1.2
1.0
0.8
0.6
0.4
0.2
0.0
Baseline p99 Spikes
Proposed p99 Spikes
UR16e
KUKA iiwa 7